



University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

3 – Input OR Gate Using Diode

Project Description

Project Name	3 - Input OR Gate Using Diode
Course Title	Basic Electronics - LAB
Submission Date	15-July-2025
Faculty	Faculty Of Information Technology
Software Used	Tinkercad



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1. Executive Summary :

This project aims to **design and implement a 3-input OR gate** using diodes on a breadboard. The goal is to simulate logical OR behavior with discrete electronic components, particularly focusing on **diodes, resistors, and a single LED output**.

- A 3-input OR gate outputs HIGH (LED ON) if **any one or more of the inputs is HIGH**. This project demonstrates how diode logic works, reinforcing key concepts of digital electronics using simple components.

2. Methodology - Components Used :

- Breadboard
- 3 × 1N4001 diode
- 3 × Slide Switch
- 3 × Resistors (1 k Ω for Switches)
- 1 × Resistor (220 Ω for LED current limiting)
- 1 × LED (output indicator)
- Jumper wires
- 9V battery

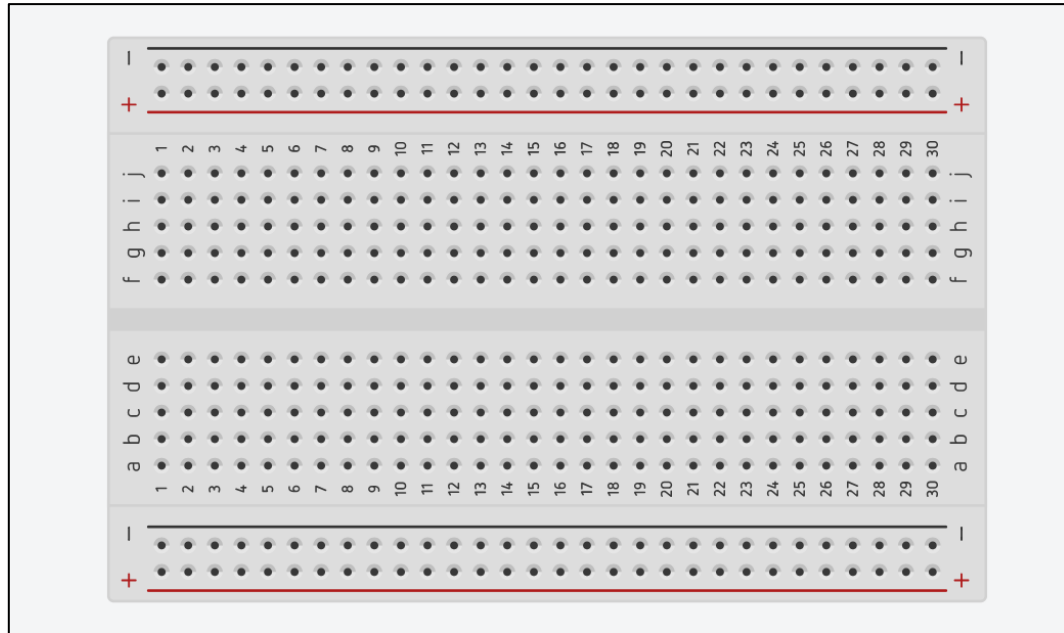


Figure 1: BreadBoard Circuit

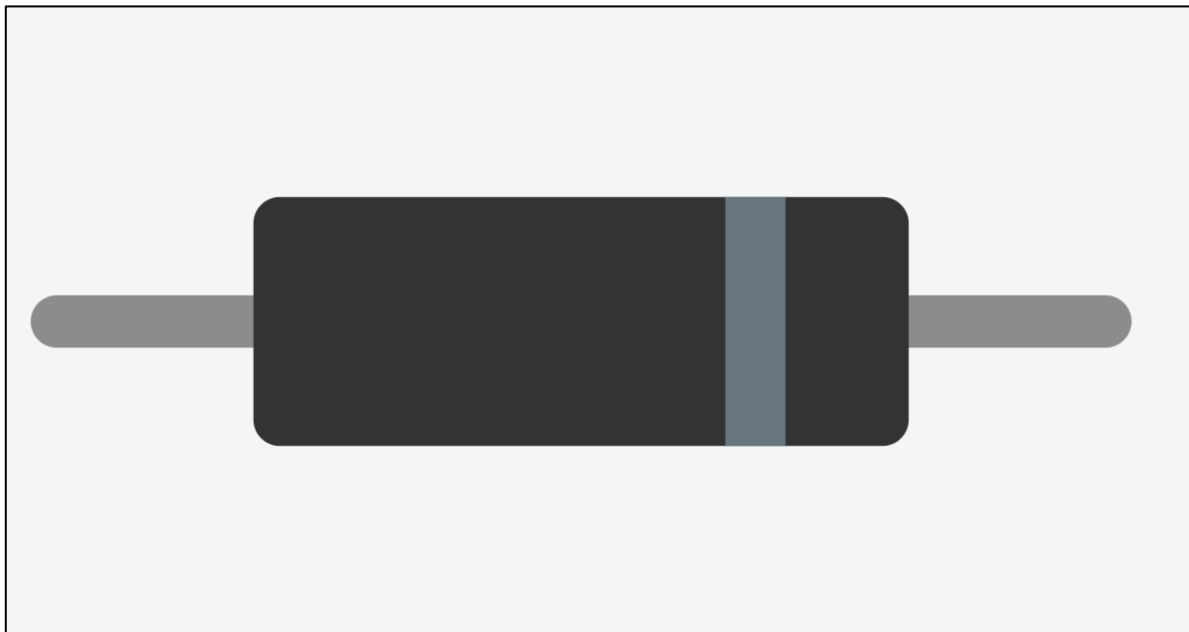


Figure 2: Diode

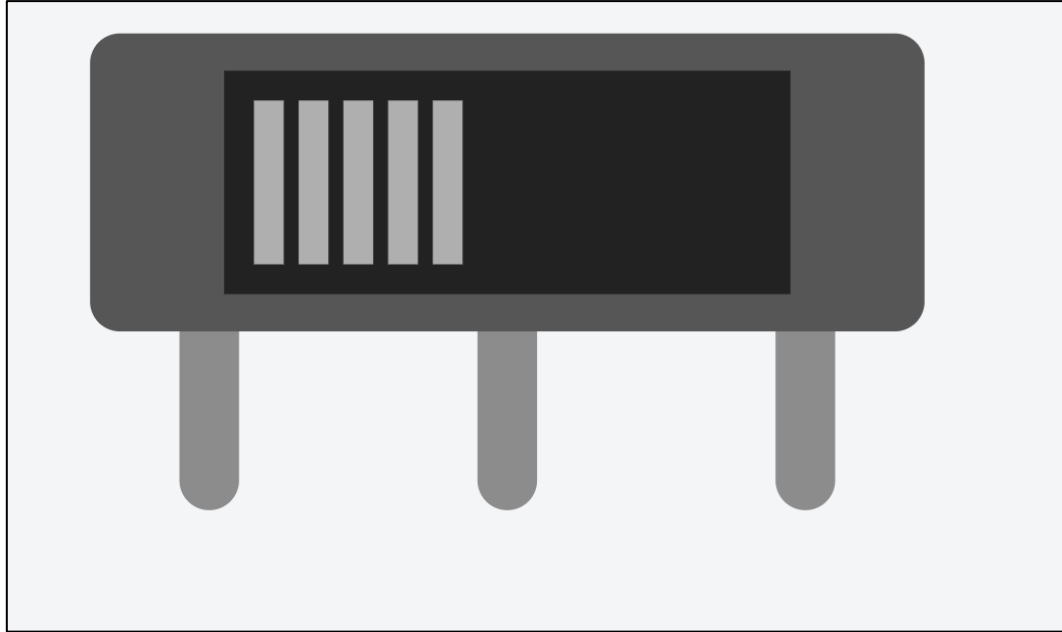


Figure 3: Sliding Switch

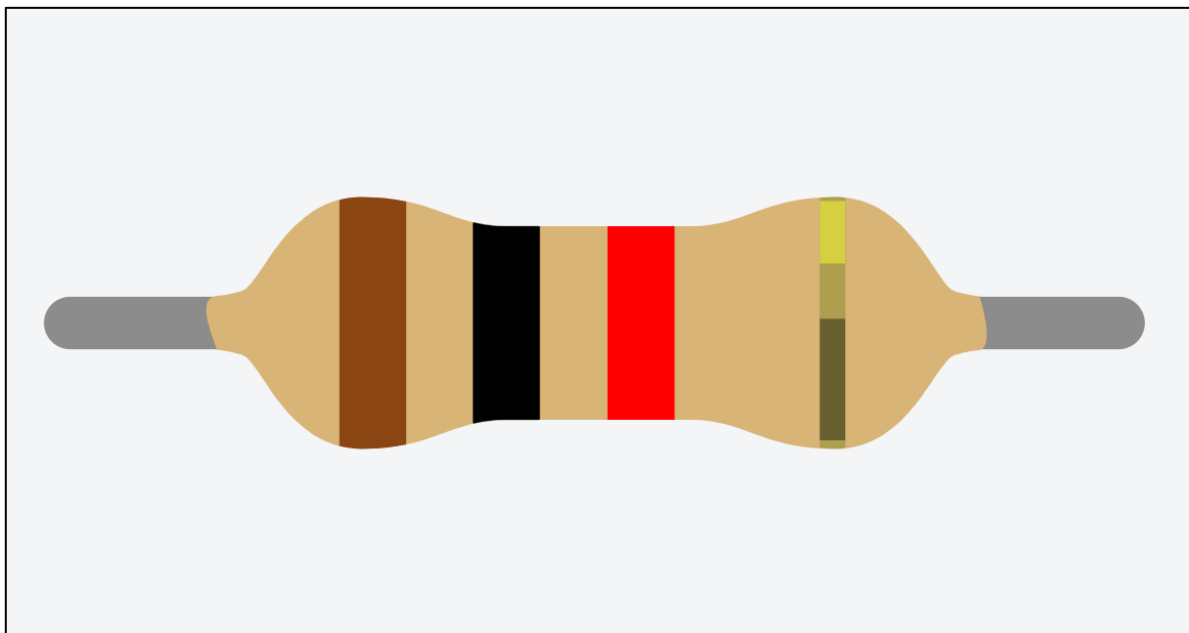


Figure 4: 1k Ohm Resistor

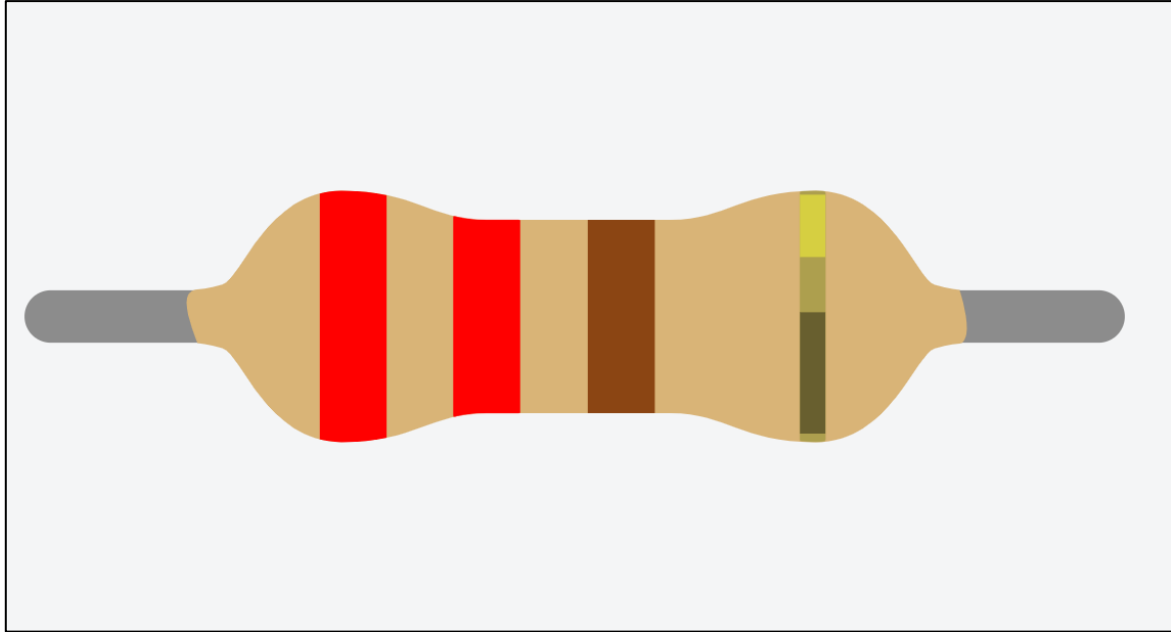


Figure 5: 220 Ohm Resistor

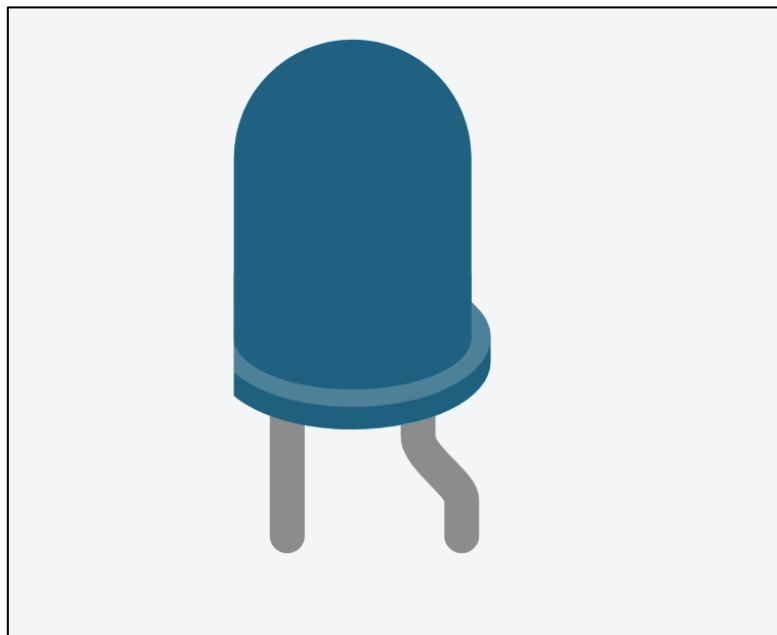


Figure 6: Blue LED Bulb



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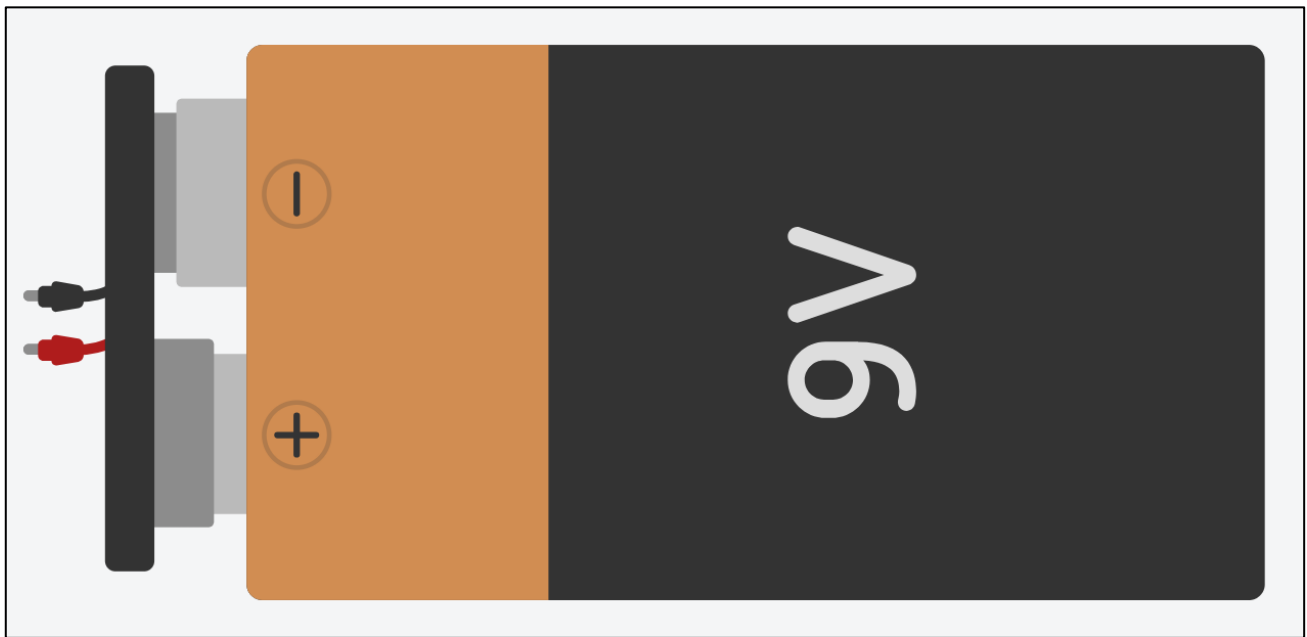


Figure 7: 9V Battery VSource

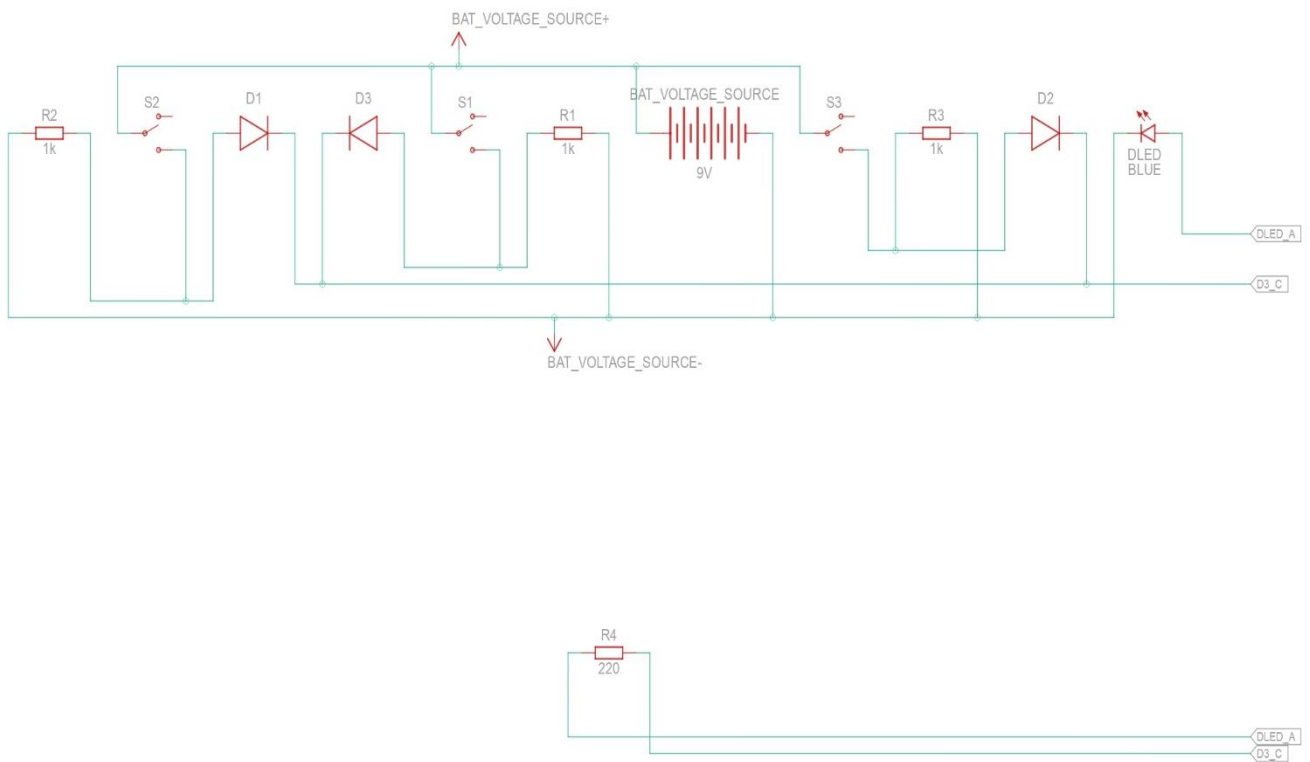


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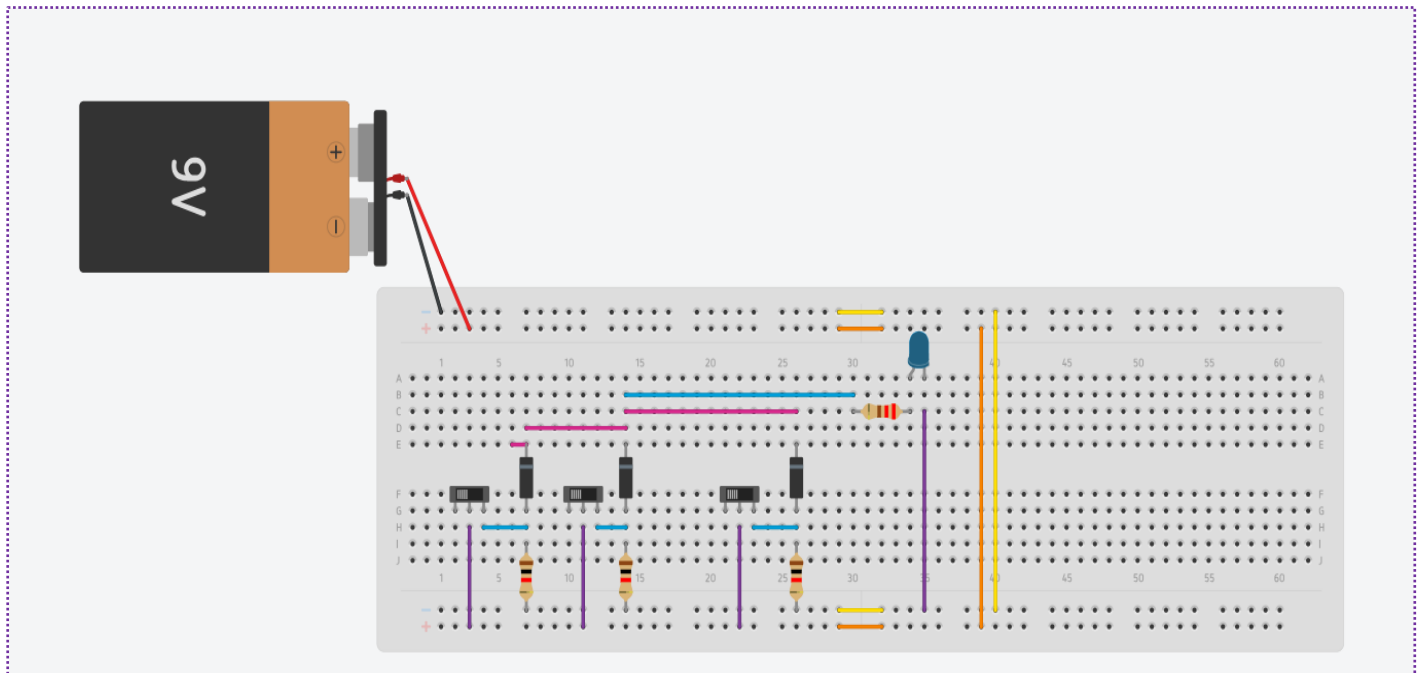
3. Circuit Diagram :



Title: 3 Input OR Gate Using Diode

15 July 2025 Final Project Basic Electronics Lab

Circuit Diagram On Tinkercad :



4. Implementation Steps:

1. **Power Setup:** Connect the 9V battery to the breadboard's power rails.
2. **Input Switch:** Connect 3 slide switch with pull-down resistors to ground and connect each to the anode of a diode.
3. **Diode Logic:** All diode cathodes are joined together and connected through a 220Ω resistor to an LED, then to ground.
4. **LED Output:** LED turns ON if any slide switch is slide to On.
5. **Verification:** Simulate or manually test the output for all input combinations.



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⚠ Challenges Faced

- Ensuring proper orientation of diodes (anode/cathode)
- Managing voltage drops across diodes (~0.7V)
- Stabilizing LED output brightness (solved using appropriate current-limiting resistor)

5. Results and Findings :

- ❑ **When all buttons are off** , the output is LOW → LED is OFF.
- ❑ **Sliding any one switches** sends a HIGH signal → LED turns ON.
- ❑ **Sliding all switches simultaneously** still results in LED ON → OR logic confirmed.

Input A	Input B	Input C	Output (LED)
0	0	0	OFF (0)
1	0	0	ON (1)
0	1	0	ON (1)
0	0	1	ON (1)
1	1	1	ON (1)